

PRYOR FIELD, AL, USA

WMO: 722279

Lat: 34.653N

Lon: 86.945W

Elev: 180

StdP: 99.18

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53852

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-5.3	-13.9	1.1	-4.8	-11.8	1.4	-3.1	10.9	10.2	9.5	10.1	2.8	350	0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	35.0	23.9	33.8	23.8	32.7	23.6	25.8	31.3	25.4	30.9	24.9	30.2	3.1	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	19.7	28.3	23.8	19.1	27.8	23.3	18.4	27.1	80.7	31.4	78.4	30.8	76.2	30.4	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.0	(b) 8.0	(c) 7.1	(e) -11.3	(f) 37.0	(g) 2.5	(h) 1.8	(i) -13.2	(j) 38.3	(k) -14.6	(l) 39.3	(m) -16.1	(n) 40.3	(o) -17.9	(p) 41.6		
(a) 9.0	(b) 8.0	(c) 7.1	(e) -11.9	(f) 27.0	(g) 2.4	(h) 0.6	(i) -13.6	(j) 27.4	(k) -15.0	(l) 27.8	(m) -16.3	(n) 28.1	(o) -18.0	(p) 28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.4	5.0	7.3	11.6	16.5	21.1	25.2	26.4	26.1	23.1	16.9	10.7	6.8
	DBStd	8.78	6.07	5.51	5.19	4.39	3.64	2.27	2.01	2.27	3.38	4.66	5.01	5.32
	HDD10.0	522	178	109	44	5	0	0	0	0	0	4	50	131
	HDD18.3	1721	413	311	214	84	18	0	0	0	6	83	232	359
	CDD10.0	2872	24	33	94	202	345	456	507	498	392	218	72	31
	CDD18.3	1028	1	1	6	31	104	206	249	239	148	39	4	1
	CDH23.3	9666	0	3	57	261	918	1983	2424	2299	1318	374	27	2
	CDH26.7	4072	0	0	7	48	308	891	1097	1077	541	101	3	0
Wind	WSAvg	2.8	3.4	3.5	3.5	3.5	2.9	2.4	2.1	1.9	2.2	2.5	2.9	3.1
Precipitation	PrecAvg	1383	140	131	134	123	106	97	126	93	89	75	106	162
	PrecMax	1746	282	261	254	250	260	205	265	205	189	181	189	457
	PrecMin	774	55	65	35	46	13	24	46	28	6	0	45	54
	PrecStd	226	53	59	52	54	66	56	47	37	55	52	40	86
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	23.4	27.5	29.9	33.0	35.7	36.2	37.4	35.1	31.6	27.1	22.4
		MCWB	17.0	17.4	17.2	19.5	22.3	23.3	24.2	23.6	22.8	21.7	18.9	17.8
	2%	DB	18.8	20.9	24.9	28.0	31.3	34.1	34.4	35.0	33.4	29.1	23.0	19.9
		MCWB	15.7	16.3	16.6	18.8	21.6	23.3	24.5	24.2	22.8	20.5	16.5	16.8
	5%	DB	17.1	18.7	22.5	26.5	29.9	32.8	33.1	33.6	31.8	27.4	21.3	17.7
		MCWB	14.4	14.4	15.5	18.3	20.9	23.0	24.5	24.0	22.5	19.0	16.2	15.2
	10%	DB	14.9	16.5	20.4	24.5	28.3	31.6	32.1	32.2	30.0	25.3	19.2	15.7
		MCWB	12.3	12.4	14.7	17.5	20.4	22.7	24.3	24.0	22.0	18.2	15.1	13.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.2	19.0	19.4	21.8	23.9	25.8	26.8	26.5	25.2	23.7	20.7	19.2
		MCDB	20.3	21.8	23.5	27.0	29.0	31.5	31.8	32.0	30.6	27.7	22.9	21.1
	2%	WB	16.6	17.7	18.0	20.5	22.9	25.1	26.1	25.7	24.4	22.5	19.2	17.4
		MCDB	18.2	20.1	22.4	25.5	28.6	31.0	31.4	31.3	29.9	27.0	21.9	19.1
	5%	WB	14.9	15.6	17.0	19.6	22.3	24.4	25.6	25.2	23.8	21.1	17.7	15.7
		MCDB	16.9	17.6	21.2	24.4	27.7	30.1	31.0	30.9	28.9	25.0	20.1	17.6
	10%	WB	12.6	13.5	15.7	18.6	21.6	23.8	25.1	24.7	23.1	19.9	16.0	13.4
		MCDB	14.3	15.8	19.7	23.1	26.6	29.1	30.3	30.0	27.6	23.5	18.5	15.2
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.8	11.9	12.2	11.6	11.5	10.8	11.5	12.4	13.2	12.5	10.2
		MCDBR	11.3	13.1	14.6	13.6	13.2	13.5	12.4	13.5	14.1	15.1	14.5	11.5
		MCWBR	8.6	8.8	7.9	6.2	5.0	4.5	4.5	4.6	5.3	6.9	8.5	8.7
	5% WB	MCDBR	9.8	10.2	11.7	11.2	10.8	11.2	10.7	11.3	11.6	11.2	10.5	10.3
		MCWBR	8.3	8.1	7.3	6.1	4.8	4.5	4.5	4.5	5.0	6.0	7.5	8.7
Clear-Sky Solar Irradiance	taub	0.322	0.327	0.369	0.401	0.449	0.480	0.521	0.530	0.432	0.380	0.341	0.319	
	taud	2.480	2.464	2.370	2.275	2.187	2.130	2.038	1.999	2.263	2.390	2.442	2.504	
	Ebn at Noon	885	920	903	886	843	813	777	761	827	847	851	865	
	Edh at Noon	86	97	115	133	147	156	170	174	127	103	88	79	
All-Sky Solar Radiation	RadAvg	2.45	2.97	4.13	5.35	5.88	6.24	5.96	5.52	4.79	3.85	2.81	2.10	
	RadStd	0.19	0.37	0.38	0.37	0.46	0.49	0.33	0.30	0.42	0.50	0.30	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (29 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+143	+102

Nomenclature: See separate page